THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest 3-digit number divisible by 9 is: 1

(a) 107 **(b)** 117 **(c)** 108 **(d)** 109

2. The largest 4-digit number with all digits different that is divisible by 9 is: 1

(a) 9873 **(b)** 9874 **(c)** 9875 **(d)** 9872

3. Kaprekar's routine on 2111 reaches 6174 after how many steps? 1

(a) 6 **(b)** 4 **(c)** 7 **(d)** 5

4. The Collatz sequence from 12 needs how many steps to reach 1? 1

(a) 11 **(b)** 9 **(c)** 10 **(d)** 8

5. For every whole number n, $n(n+1)$ is: 1

(a) odd **(b)** a multiple of 3 **(c)** even **(d)** a perfect square

6. If $1+2+3+\cdots+n=231$, then n is: 1

(a) 23 **(b)** 22 **(c)** 20 **(d)** 21

7. The number of reverse-and-add steps for 59 to become a palindrome is: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 2

8. Which expression is a multiple of 9 for every digit pair (a, b) with $a > b$? 1

(a) $(10a+b)+(10b+a)$ **(b)** $(10a+b)-(10b+a)$ **(c)** $a+b$ **(d)** $10a+b$

9. **Assertion (A):** The sum of two odd numbers is even. 1

Reason (R): An odd number is of the form 2n + 1, so the sum of two odd numbers is 2(m + n + 1).

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** 5832 is divisible by 9. 1

Reason (R): 5832 is an even number.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A two-digit number added to its reverse gives 110 and subtracted from it gives 36. Find the number. **2**

12. Show that $n(n+1)$ is even, and verify for $n = 13$. **2**

13. Using each of the digits $7, 3, 4, 1$ once, make the largest 4-digit number divisible by 3. **2**

14. A two-digit number added to its reverse gives 143 and subtracted from it gives 45. Find the number. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The number $4 * 3 * 6$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. **3**

16. Start with 1235 and apply Kaprekar's routine until 6174 appears. Show every step. **3**

17. In how many ways can 15 be written as the sum of two or more consecutive positive whole numbers? List them. **3**

OR

The number $7 * 2 * 1$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. **3**

18. Start with 1259 and apply Kaprekar's routine until 6174 appears. Show every step. **3**

OR

In how many ways can 21 be written as the sum of two or more consecutive positive whole numbers? List them. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Teacher Neha shows how 15 can be written as a sum of consecutive numbers: $15 = 7 + 8 = 4 + 5 + 6 = 1 + 2 + 3 + 4 + 5$. **5**

(i) Which of these sums uses exactly three numbers?

(ii) Write 21 as the sum of three consecutive numbers.

(iii) Is the sum of three consecutive numbers always divisible by 3? Give the reason.

(iv) Write 33 as the sum of three consecutive numbers and also as the sum of two consecutive numbers.

20. Use reverse-and-add on 155 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. **5**

OR

Find all two-digit numbers that are exactly 3 times the sum of their digits. **5**

END OF QUESTION PAPER — Total: 40 Marks